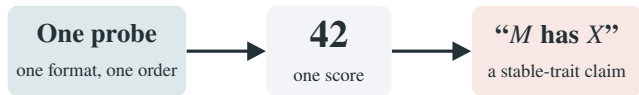


Can We Believe What Large Language Models Do?

A Survey of Validity Threats in Behavioral Studies



**What must be true
before we can make
this claim?**

The motivating puzzle: the conclusion can move while the task stays fixed

A common inferential shortcut

Observe an output $\rightarrow$ compute a score $\rightarrow$
infer a stable trait, preference, strategy,
value, or bias.

The missing question

Is the score a property of the model, or
of the probe, scoring rule, population, and
setting used to produce it?

76 points

accuracy movement after a format change

82.5%

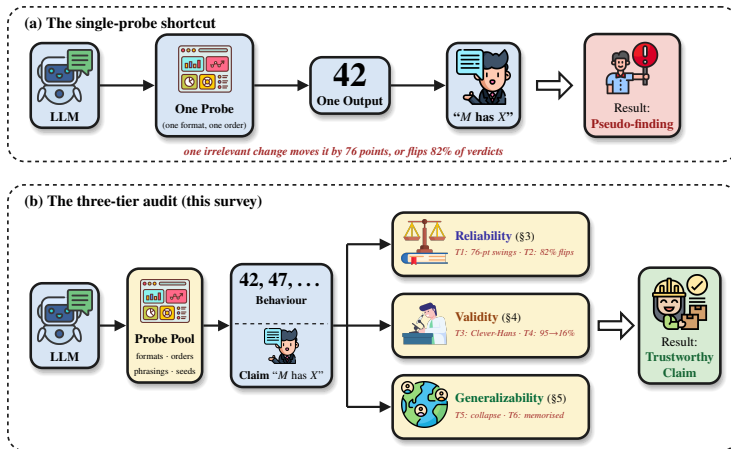
one pairwise-judge comparison flipped by order

95% $\rightarrow$ 16%

cooperation changed across model variants in
one game

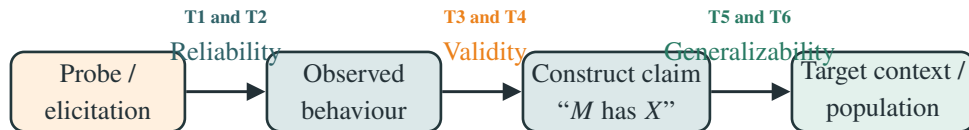
Source: Sclar et al. (2024); Wang et al. (2024); Li and Shirado (2025). These are documented extreme settings, not prevalence estimates.

Our thesis: behavioural evaluation is a measurement problem



The contribution is the audit between observation and interpretation.

One pipeline, three questions



Does the number hold still?
under irrelevant probe changes

Does it measure X ?
rather than a shortcut or scoring
choice

Does the claim transfer?
beyond the tested sample and
setting

The survey reorganises a fragmented literature into six threats

Tier 1: Reliability

T1 Prompt & format

T2 Order & position

Tier 2: Validity

T3 Construct validity

T4 Benchmark validity

Tier 3: Generalizability

T5 Population & aggregation

T6 Ecology & leakage

Same failure modes, different names: machine psychology, strategic games, social simulation, and LLM-as-a-judge.

Source: Structured narrative review of 278 cited records; selection and scope protocol reported in the appendix.

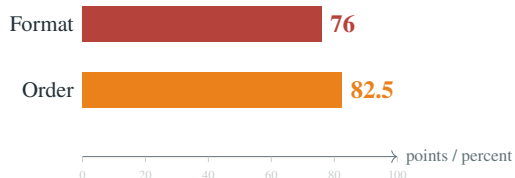
Tier 1: a stable-looking score can be a property of the probe

T1: Prompt & format

Equivalent formats, punctuation, wording, or decoding choices can change measured performance.

T2: Order & position

Models and judges can prefer a slot, label, or earlier answer independently of content.



Takeaway

One probe estimates one conditional response. It does not identify a stable model property without a declared probe pool.

Source: Sclar et al. (2024); Wang et al. (2024); Dominguez-Olmedo et al. (2024).

Tier 2: even a stable number may measure the wrong thing

Construct validity (T3)

- ▶ Human questionnaires do not automatically retain their meaning on models.
- ▶ Response style can be recorded as personality or value variance.
- ▶ Across 56 models, 81 to 90% of profile variation was directional response bias.

Normative-benchmark validity (T4)

- ▶ “Strategic”, “rational”, “human-like”, and “coherent” are different targets.
- ▶ A model may violate equilibrium play while satisfying consistent-choice axioms.
- ▶ Reporting the scoring rule fixes what the result means.

Stable numbers attach first to a test and a scoring standard, not automatically to a construct.

Source: Gupta et al. (2024); Shu et al. (2024); Meyer et al. (2026); Akata et al. (2025); Chen et al. (2023).

Tier 3: when the probe does not resemble deployment

> 0.99

probability on one modal opinion
after distribution collapse

OpinionQA

50.5% / 22.8%

strongest model mean / hardest
split under information asymmetry

InformativeBench

7% → 80%

welfare after a payoff rewrite in one
social-dilemma setting

CoopEval

T5: population & aggregation

Means can flatten subgroups, interactions, and
human variability.

T6: ecology & leakage

Canonical or memorised probes can overstate
transfer to deployment.

Source: Santurkar et al. (2023); Liu et al. (2024); Tewolde et al. (2026). Each magnitude is tied to its reported setting.

Contribution 2: quantify fragility relative to signal

$$\phi = \frac{V_{\text{probe}}}{V_{\text{total}}}$$

V_{probe} : variation caused
by a declared probe axis

V_{total} : all measured variation

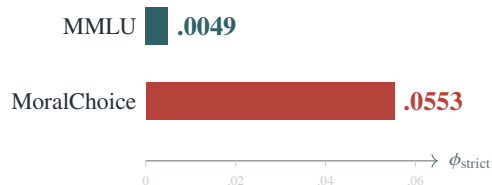
Strict ϕ uses the probe main effect; an interaction-aware version adds item-by-probe variation.

Why not report sensitivity alone?

- ▶ The same prompt movement matters differently when model separation is large or tiny.
- ▶ ϕ is claim-, model-, and axis-specific.
- ▶ It converts stability into a design question: how many probes are needed?

A small ϕ validates only the declared axis, not the construct.

Matched comparison: the tested axis accounts for 11× more variation



Matched design

- ▶ Item-level analysis with difficulty matching.
- ▶ Same estimator, explicit probe axis.
- ▶ Eleven-fold contrast between these measurements.

Not supported: “behaviour is generally eleven times more fragile than capability.”

Supported claim: this dilemma measurement depends more on its tested variations than the matched capability measurement does.

We tried to falsify our own headline

Positive control

A behavioural positive control with a strong expected preference has median $\phi_{\text{strict}} = 0.0005$ under language variation.

Behaviour is not inherently fragile.

Axis dependence

The same control reaches $\phi = 0.30$ when presentation order is added.

Robustness on one axis does not transfer to another.

Ceiling check

Easy capability items have little binary variance left to move.

Some apparent robustness is arithmetic, not substantive.

**Fragility depends on three things
(claim, probe axis, model).**

Uncertainty changes what we are allowed to say

.034 $\rightarrow$.172

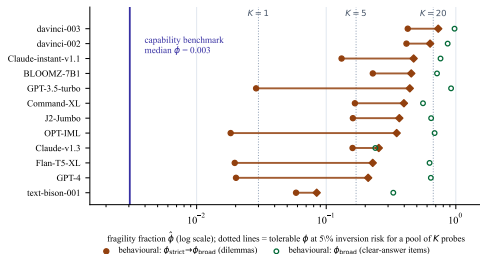
median interval width after resampling probes as

well as items
3 / 66

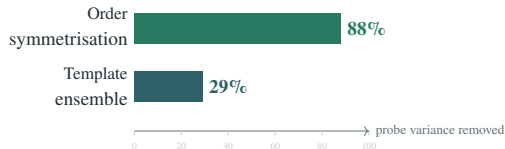
model pairs that remain separated under the paired
bootstrap

Implication

Point rankings are descriptive. Pairwise superiority
claims need paired uncertainty over the probe pool.



The metric leads directly to a better intervention



A practical rule

1. Decompose variation by probe axis.
2. Target the term that carries the variance.
3. Re-estimate uncertainty after mitigation.

On clear-answer items, the ordering reverses. The best remedy is empirical, not universal.

**The decomposition explains why the remedy worked:
answer order carried most of the measured instability.**

End-to-end audit: what can we still claim?

Claim that weakens

“The model has a stable personality profile.”

Equivalent prompts move scores; response style dominates between-model variance; self-report weakly predicts open-ended behaviour.

Comparative claim that survives

On the tested dilemma probes, GPT-4 is less probe-dependent than the other eleven models.

The claim is relational and explicitly scoped to the probe pool.

The audit does not force nihilism. It converts an over-broad trait claim into the strongest claim the evidence supports.

A checklist authors and reviewers can run

Design

1. Name the construct and target context.
2. Declare a probe pool and expected invariances.
3. Balance order; vary formats, wording, and seeds.
4. Include an ecological or subgroup stress test.

Analysis & reporting

5. Decompose model, item, probe, and interaction variance.
6. Bootstrap items and probes together.
7. Validate the test and state the scoring standard.
8. Report the claim with its tested scope.

Release probe-level outcomes: without them, fragility cannot be re-audited.

What changes if the field adopts this view?

For benchmark builders

A benchmark becomes a sampled measurement design, not a single prompt template.

For reviewers

Ask which link supports the claim, which axis was varied, and what remains invariant.

For deployment

Prefer scoped comparative claims over essentialist statements about what a model “is”.

Impact = fewer pseudo-findings, clearer scopes, and interventions targeted at measured failure modes.

One output can support “**the model produced this response**”.
It does not by itself support “**the model has this stable trait**”.

1. Audit the link

reliability → validity → transfer

2. Measure fragility

relative to the signal and named
axis

3. Scope the claim

say exactly what survives the audit

Questions? Repository / QR code placeholder: `anonymous-url`

Backup: review scope and evidence policy

- ▶ Structured narrative review, not a PRISMA systematic review.
- ▶ 278 cited records across games, social simulation, machine psychology, and LLM-as-a-judge.
- ▶ Headline magnitudes are tagged by role: extreme stress test, broad evidence, mechanism, or mitigation.
- ▶ Preprints support the map but do not carry a threat's headline alone.
- ▶ Main evidence is densest in forced-choice English text; multilingual, multimodal, and open-ended evidence extends the taxonomy without proving that every threat grows outside that setting.

Backup: variance decomposition behind ϕ

For one model, outcome y_{ik} on item i under test variant k is

$$y_{ik} = \mu + a_i + b_k + (ab)_{ik} + \varepsilon_{ik}.$$

The strict version counts only the shared probe shift. The broad version also counts cases where the same probe helps some items and hurts others:

$$\phi_{\text{strict}} = \frac{\sigma_b^2}{\sigma_{\text{total}}^2}, \quad \phi_{\text{broad}} = \frac{\sigma_b^2 + \sigma_{ab}^2}{\sigma_{\text{total}}^2}.$$

Broad ϕ requires repeated observations to separate interaction from run-to-run noise. With one binary observation per cell, only strict ϕ is identifiable.

A directional probe main effect may be obvious in the mean,
while an item-by-probe interaction can silently scramble rankings.

Backup: using ϕ to size the probe pool

Probe noise is shared within a probe and therefore is not averaged away by adding more items or samples. More independent probes are required. At 5% accepted inversion risk in the MoralChoice-anchored independence case:

Probe count K	Permitted ϕ
1	0.03
5	0.17
10	0.34
20	0.67

A universal cutoff is not meaningful without K , the model gap, and an accepted error rate.

Backup: remedy cost and what each intervention buys

Remedy	What it addresses	Main cost
Balanced permutation	Signed order / position effects	2× for pairwise judging
Probe or template pool	Format, wording, seed variation	Linear in pool size K
Variance decomposition	Locates the dominant axis	No new inference if outcomes retained
Gold-slice calibration	Corrects surrogate labels; returns coverage	Human-labelled subset
Ecological stress test	Measures transfer rather than assuming it	Task-specific environment

Backup: sources for headline examples

Example	Primary source
76-point format swing	Sclar et al. (2024), <i>Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design</i>
82.5% order conflict	Wang et al. (2024), <i>Large Language Models are not Fair Evaluators</i>
Opinion distributions	Santurkar et al. (2023), <i>Whose Opinions Do Language Models Reflect?</i>
Information asymmetry	Liu et al. (2024), <i>Autonomous Agents for Collaborative Task under Information Asymmetry</i>
Cooperation rewrite	Tewolde et al. (2026), <i>CoopEval</i>
Response-style variance	Meyer et al. (2026), <i>Apparent Personality in Language Models</i>